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A survey of energy efficient methods for UAV communication

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ABSTRACT

Unmanned aerial vehicles (UAVs) are characterized by high flexibility and low deployment costs. Therefore, UAVs, in recent years, have been widely used in all walks of life, such as logistics services, search and rescue, agriculture and mining. These applications are highly dependent on UAV communication technology. However, the UAVs currently in use are still powered by low-power batteries, which greatly limits the operational life of UAVs, resulting in varying degrees of communication interruptions and cost increases. Therefore, improving the energy efficiency (EE) of UAV communication has become a key problem that needs to be urgently solved. This article gives a comprehensive review of methods to improve the EE of UAVs through UAV trajectory planning and deployment, resource allocation and management, energy saving communication protocol design and UAV energy harvesting (EH) and transfer. Additionally, this paper engages with related research to put forward some promising research directions.

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1. Introduction

In recent years, with the advance of science and technology, the UAV industry has also achieved rapid development, and has entered the golden age of commercial application of UAV. According to the data from the China Industrial Research Institute, the number of UAVs around the world exceeded 4 million by the end of 2020, and the number will increase exponentially in the next few years. Due to the high flexibility and low cost of UAVs, UAVs can be widely used in emergency response [1], logistics services [2], search and rescue [3], agriculture [4], mining [5] and other fields. A UAV cannot successfully complete these tasks without the guaranteed reliability of UAV communication technology [6]. The commercialization of the fifth generation networks (5G) technology provides UAV communication with ultra-high speed, very low latency, high data rate transmissions and other capabilities, which effectively solves the problem of UAV communication quality [7]. However, UAV communication still faces many optimization problems. These problems include, UAV aerial base station (BS) deployments, spectrum utilization, limited battery lives, privacy security and legal and regulatory issues [8].

Among the above issues, the energy consumption of UAV communication is a key problem restricting the network lifetime. Therefore, in recent years, the research on improving the EE of UAV communication system has become more and more extensively.

Based on the relevant data, the longest duration time of commercial UAVs has reached 80 hours, 46 minutes and 35 seconds [9]. However, UAVs are battery-powered devices that use low-power batteries, and their payload is limited by the size and capacity of their battery. In addition to accomplishing a specific task, a UAV still needs energy to communicate and fly [10], therefore the energy supplies are finite. Because of the limited energy of the UAV, communication interruptions and cost increases will inevitably occur in UAV communication systems.

Communication interruption. The UAV needs to be charged when the battery is exhausted, which inevitably leads to communication interruptions [11]. The age of the sixth generation networks (6G) is coming, and the goal of 6G is to provide a space-air-ground-underwater network with ubiquitous coverage. UAVs play an critical role in 6G communication and can assist in the deployment of maritime communication BSs [12]. Therefore, to avoid communication interruption and improve the reliability of the network systems, resolving the energy limitations of UAVs and improving their endurance are key problems to be urgently solved for UAV communication [13].

Cost increase. At present, when the power of a UAV runs out, the battery of the UAV is replaced or the failing UAV is substituted

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Table 1
Relevant surveys on EE of UAV communication.

Survey	Year	Brief summary	Difference from our work
[12]	2022	A survey of energy consumption models, typical applications and development trends of green UAV communication in 6G	Only three energy efficient methods including trajectory planning, power allocation and energy harvesting are researched in this paper, **while** energy efficient communication protocols are also proposed in our paper except the above three methods
[14]	2021	A survey of the problems, challenges and common technologies of UAV communication	Only a kind of energy efficient methods of communication protocol is researched in this paper, **while** there are three other methods are also proposed in our paper except the above method
[15]	2021	A brief survey on UAV communication combined with MEC	Only a kind of energy efficient methods of resource allocation is researched in this paper, **while** there are three other methods are also proposed in our paper except the above method.
[16]	2019	A survey of public safety of UAV communication from the perspective of energy saving	Only the public safety issue of UAV communication from the energy efficient perspective is researched in this paper, **while** there are four energy efficient methods in UAV communication are researched in our paper
[17]	2022	A review of energy optimization algorithms based on UAV cellular networks	Only a kind of energy efficient algorithms of the deployment of UAVs is researched in this paper, **while** there are three other methods are also proposed in our paper except the above method
[18]	2020	A review of data delivery models and routing protocols in UAV communication	Only a kind of energy efficient methods of communication protocol is researched in this paper, **while** there are three other methods are also proposed in our paper except the above method
[19]	2018	A review of the energy consumption model and influencing factors of UAV routing	Only energy consumption models and impact factors are researched in this paper, **while** the specific energy consumption optimization methods are also proposed in our paper except the above content

with a fully powered one. However, considering the increasing use of UAV communications, these approaches would undoubtedly require a large number of spare batteries or UAVs, which would lead to unacceptable operating costs.

1.1. Relevant surveys and tutorials

In order to avert the above crunch, various energy efficient approaches have been proposed, while a couple of surveys and tutorials have also been published to summary and analysis them, which are outlined at a glance in Table 1.

More specifically, authors in [12] have reviewed the green UAV communication synthetically including energy efficient models, techniques and application of UAV communication. They highlighted the critical need for EE in UAV communication from the perspective of 6G and battery limitation of UAV. Green techniques concluded into five categories, communication protocol has not been involved. However, green techniques as one of the part of [12] have not been discussed elaborately. Furthermore, neither of the outstanding feature, advantages, and limitation of these green techniques were reviewed directly. The work in [14] can be regarded as a supplement for [12], where the performance comparison of energy efficient communication method has been made solely. Especially energy efficient routing protocol in UAV was regarded as one of the important methods for UAV communication in the past five years before the publication of [14]. The parameters and metrics utilized in the method were analyzed clearly. However, green methods in [14] are too simple caused by the limitation of time.

Authors in [15,16] research energy efficient UAV communication method under certain scenarios. A brief survey on UAV communication combined with mobile edge computing (MEC) is presented in [15]. The overall network architecture, access schemes, the types of optimization problem and solution have been concluded. Resource allocation was regarded as a common method for this scenario. Public safety has been considered with UAV communication from the energy efficient perspective in [16], where a multi-layered energy efficient network architecture was proposed. They emphasized that numerous factors affecting UAVs energy consumption including UAV placement, flight time, mobility pattern, etc. The factors of the three ones are analyzed separately. Actually, the multi-layered energy efficient network architecture can be concluded as the energy efficient method of UAV placement.

Authors in [17–19] all give a detailed introduction of a single algorithm. In [17], energy optimization techniques were concluded by two types-conventional and machine learning algorithms. The state-of-the art and current trend were presented clearly with these two types. Novel techniques that empower UAV communication were also considered. However, the comparison of convention and new algorithm has not been given in [17]. Works in [14,18] both provided the survey on energy efficient routing protocols. In contrast to the content in [19], the energy efficient routing protocols were classified by the types of routes, kinds of protocols in [18]. While the influence of different factors on UAV routing were emphasized in [19]. Energy consumption models for UAVs with different factors were also given in [18].

Different from the above surveys, existing energy efficient methods of UAV communication are summarized and classified in this paper. A detailed introduction of the main idea, outstanding feature, advantages and application scenario of each method was made elaborately.

1.2. Energy efficient methods for UAV communication

Existing methods for increasing EE of UAV communication were usually designed from two ways: one is to reduce the energy consumption, another one is to harvest the energy [8]. Reducing communication energy means minimizing the transmission power or improving communication EE by developing optimal transmission schedule. Trajectory and deployment of UAV, transmission resource are all the conditions that affect the above optimization problems. Based on their features, energy efficient methods were grouped under four broad categories as follows in this paper.

Energy efficiency in trajectory planning and deployment. To design an energy saving scheme for UAV communication, the first approach is UAV trajectory planning and deployment. The UAV requires additional propulsion energy consumption to maintain flights at high altitudes with the necessary maneuverability, which requires much more energy than the communication energy consumption. The propulsion energy consumption of a UAV is determined by its flight status. Therefore, the optimization of UAV trajectory planning and deployment can greatly improve the EE of UAV communication.

Energy efficiency in resource allocation and management. The second method is to rationally allocate the spectrum resources, power resources and computing resources of the UAV communi-

cation system. The coverage and monitoring range of a single UAV is limited in the process of mission execution, and multiple UAV groups often coordinate to achieve coverage integrity. Therefore, making rational use of these resources will effectively improve the EE of UAVs [20].

Energy efficiency in communication protocol design. The third method mainly focuses on UAV ad hoc network and reduces the energy consumption of data transmissions in UAV ad hoc network by energy-saving designs of the relevant communication protocols at the physical layer, data link layer and network layer.

Energy efficiency in energy harvesting and transfer. The fourth method is to harvest the energy directly on the working UAVs. For example, UAVs can be powered by energy harvested from solar or wind, or they can use radio frequency (RF) energy and laser beam energy to directly supplement the energy for the UAVs by using wireless charging technology [21].

1.3. Motivation and contributions

As we discussed above, existing surveys on energy efficient UAV communication only give a detailed introduction to a particular EE method or limited by a certain application scenario. Classification and comparison between different methods are needed to show the overall design issues for EE in UAV communication. We attempt to summarize different methods of EE improvement for UAV communication in recent years with three main objectives:

- To highlight the significance of EE in UAV communication.
- To summary and analysis the different EE methods in UAV communication, provide the outstanding features, disadvantage, and advantage of them in order to understand where we currently stand and the problem we across.
- To identify the problems we across, based on which, to summarize the further research should be focused on.

To achieve the above main objectives, the contributions on our paper are as follows:

- The energy consumption models of different UAVs is concluded in this paper, based on which, various trajectory optimization and the deployment of UAV are concluded. The main idea, outstanding features, and limitation of trajectory optimization are analyzed detailedly. These analyses make a comparison clearly and intuitively for readers.
- Different from other works where network resources are only considered, calculating resource allocation is also discussed with considering network latency. This reveals that the design of energy efficient methods need consider other performance of networks, since 6G and future networks will be the networks that meet multiple requirements simultaneously.
- Energy efficient communication protocol design is a new method we conclude based on the original routing protocol of UAV communication. The introduction of it includes the protocol in the layer of network and data link. The advantages of different protocols are also contrasted clearly.
- EH technologies are also summarized of renewable energy and electromagnetic radiation. Based on the analysis and summary of the existing methods, other technologies can boost EE and future research directions are shown in the part of future research challenges.

1.4. Organization of the paper

As shown in Fig. 1, this paper is mainly developed according to the above four energy efficient methods, and the four methods are elaborated separately in sections 3–6. And Table 2 shows a summary of main acronymns and their definitions used throughout this paper. The specific structure is as follows: Section 2 introduced the possible architectures in energy efficient UAV communication system and given the EE definition of UAV communication. Section 3 summarizes the first method for EE of UAV trajectory planning and deployment. Resource allocation and management for EE maximization is introduced in Section 4. Section 5 sums up the energy saving communication protocol in terms of network layer and data link layer. Different from the above, which maximized the EE of the system itself. Section 6 presents the energy harvesting technology for UAV communication to enhance the EE with extra energy. Section 7 shows the possible future research directions based on existing problems. Lastly, Section 8 concludes the survey with final remarks.

2. Architectures in UAV communication

2.1. Classification of UAVs

UAV is a significant part of the UAV communication system. Different scenarios apply to different classification of UAVs. According to the airfoil construction of UAVs, they are also divided into two categories. One is fixed-wing UAVs, and the other is rotary-wing UAVs [22].

Fixed-wing UAV. Fixed-wing UAVs use thrust from propellers or turbine engines to power the UAV to fly forward. The advantages of fixed-wing UAVs are that the load capacity of them is large and the flight speed is fast. But there are also some limitations that a fixed runway is required when takeoff and landing.

Rotary-wing UAV. Rotary-wing UAVs rely on lift generated by multiple rotors to balance the gravity of the vehicle, allowing the vehicle to fly and control the stability and attitude of the vehicle by changing the speed of each rotor. The advantages of rotary-wing UAVs are that neither taking-off nor landing requires a fixed runway and they can hover in the air. But the limitations are that the load capacity is smaller than fixed-wing UAVs and the flight speed is slower. The characteristics of the two kinds of UAVs are compared in Table 3.

2.2. Possible architectures in UAV communication

According to the application of UAVs, UAV communication can be divided into two forms: UAV-assisted wireless communication and cellular-connected UAV communication. The types of UAV communication are shown in Fig. 2.

UAV assists wireless communication. Dedicated UAVs are employed as aerial BSs, relays, or access points (APs), to assist the wireless communications of ground nodes. Resource allocation, energy harvesting and communication protocol are the more common types of energy efficient methods for UAV communication. Energy efficient communication protocol is more involved in the joint networking of multiple UAVs. With consideration the mobility of UAVs, trajectory planning and deployment for UAVs is regarded as an energy efficient method for UAV communication. The technology of energy harvesting for environmental energy is important due to the limited battery. This architecture is applicable in areas where coverage of cellular network is poor, or in areas of Internet of Things (IoT) communication.

Cellular-connected UAV communication. UAVs for various applications are integrated into the existing and future cellular networks as new aerial user equipment. Different from UAV-assisted wireless communication, Energy efficient resource allocation need consider the coexistence of cellular users with cellular spectrum resource. Since UAV communicate directly to the BS, the design

Table 2
Main acronyms and their definitions.

Acronyms	Definition
2D	Two dimensional
3D	Three dimensional
3GPP	The third generation partnership project
5G	The fifth generation networks
6G	The sixth generation networks
ACO	Ant colony optimization
AODV	Ad hoc on demand distance vector
APs	Access points
BCD	Block coordinate descent
BS	Base station
CGW	Cluster gateway
CH	Cluster head
CMs	Cluster members
CPP	Covering path planning algorithm
CREEP	Cluster-head restricted energy efficient protocol
CS	Cuckoo search algorithm
D2D	Device-to-device communication
DAI	Decomposition and iteration
DCA	Difference of convex algorithm
DE	Differential evolutionary
DRL	Deep reinforcement learning
DP	Dynamic programming
DSR	Dynamic source routing
EA	Evolutionary algorithm
EALC	Energy aware link-based clustering
ED-LAR	Energy and direction aware location aided routing protocol
EE	Energy efficiency
EECRED	Enhanced energy conservation based on residual energy and distance
EF-MAC	Energy-efficient and fast multiple access control
EFC-TDBA	Energy-efficient fair communication through trajectory design and band allocation
EH	Energy harvesting
EH-UAV-CRN	UAV-based cognitive radio networks with energy harvesting
EL-MAC	Energy-efficient and location-aware medium access control
EM	Electromagnetic
EMDR	Energy-efficient multihop data routing
ETSI	European telecommunication standards institution
FANETs	Flying ad hoc networks
FSO	Free space optics
GAs	Genetic algorithms
GFR	Flooding-based routing protocol
GRP	Geographic routing protocol
LAA	Licensed-assisted access
LEACH	Low-energy adaptive clustering hierarchy
LoS	Line-of-sight
IoT	Internet of things
LSTM	Long short-term memory
LTE	Long-term evolution
MAC	Multiple access control protocol
MEC	Mobile edge computing
MIMO	Multiple-input-multiple-output
MO-OLSR	Multiobjective optimized link-state routing protocol
MORP	The multiobjective routing protocol
NOMA	Non-orthogonal multiple access
OFDMA	Orthogonal frequency division multiple access
OFM	Orthogonal multiple access technology
OLSR	Optimized link state routing
PDP	Pickup and delivery problem
PECM	Propulsion energy consumption
PSO	Particle swarm optimization
PSO-GLER	An ad hoc routing protocol for FANETs based on greedy for warding and limited flooding
PSR	Right division protocol
QoE	Quality of experience
QoS	Quality of service
RIS	Reconfigurable intelligent surface
RF	Radio frequency
SA	Simulated annealing
SCA	Successive convex approximation
SE	Spectrum efficiency
SEMRP	A swarm energy-efficient multicast routing protocol
SNR	Signal-to-noise-ratio
TPA	Trajectory planning algorithm
TSP	Traveling salesman problem
TSR	Time exchange protocol
UAV	Unmanned aerial vehicle
UADG	UAV-assisted data gathering
UE	User equipment
UF	Utility function
WOA-OLSR	A whale optimization algorithm based on optimized link state routing
WSN	Wireless sensor network

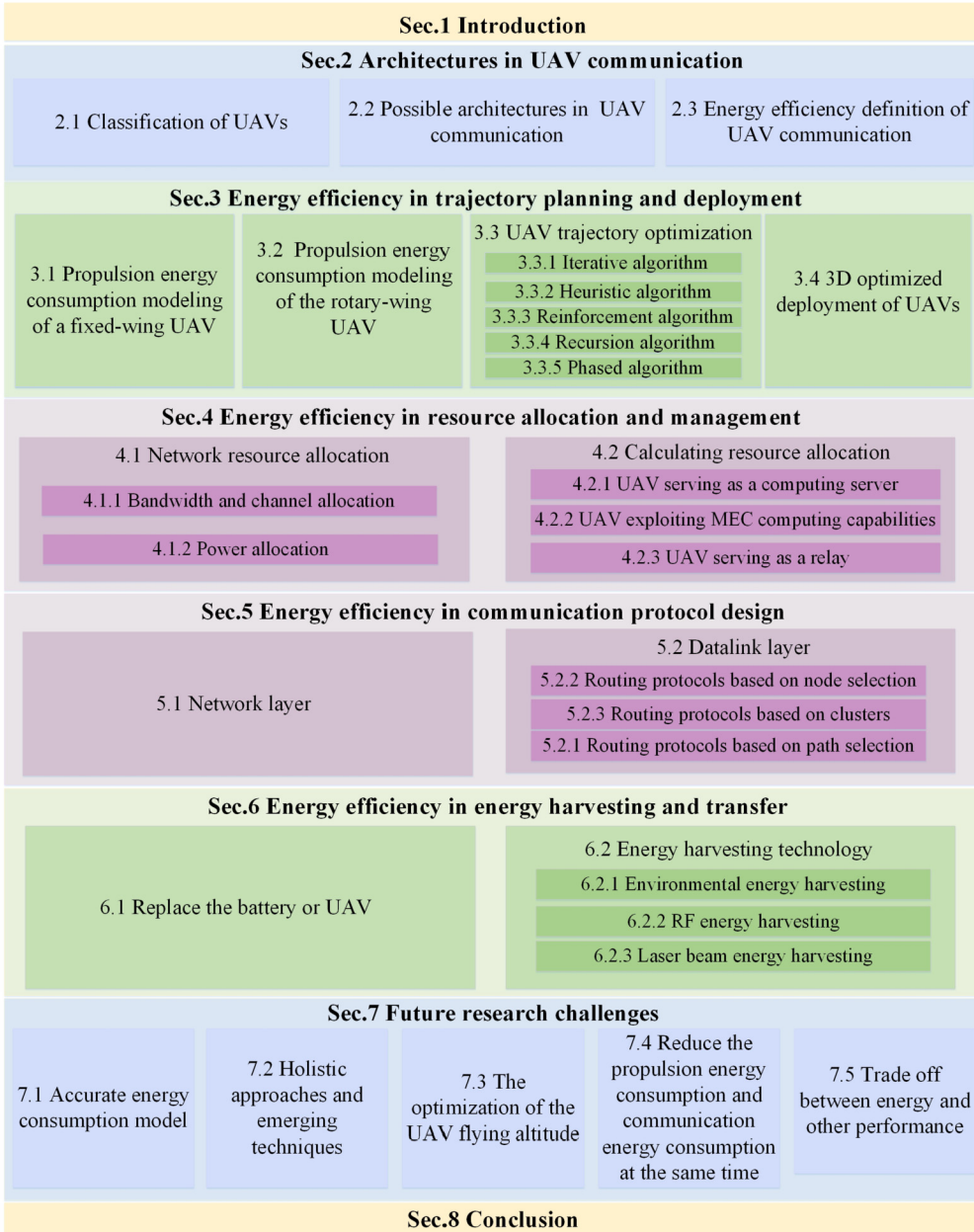

Fig. 1. Article structure.

Table 3
UAV classification.

Type	Payload capacity	Take-off and landing	Can hover
Fixed-wing UAV	Large (≤200 kg)	Need a fixed runway	No
Rotary-wing UAV	Small (≤15 kg)	No need for a fixed runway	Can

of energy efficient communication protocol is relatively rare. Compared to UAV-assisted wireless communication, the technology of energy harvesting in cellular-connected UAV communication for RF energy harvesting is more important due to the high transmission power of BS. This architecture is often applicable in hotspot areas to alleviate the overload of BS.

2.3. Energy efficiency definition of UAV communication

To implement an energy saving design for UAV communication systems, at first, the energy consumption and EE of the system needs to be modeled, and the source of energy consumption and the key factors affecting energy consumption need to be clarified so that the energy consumption of the system can be reduced in a targeted manner.

According to [8], the energy consumption of the UAV is mainly composed of propulsion energy and communication energy. Propulsion energy refers to the energy consumed on the take-off, landing, hovering and the high altitude flying of the UAV. Communication energy includes the energy consumed in the calculating, sending and receiving of information [23]. Therefore, the energy consumption E of the UAV system is expressed as:

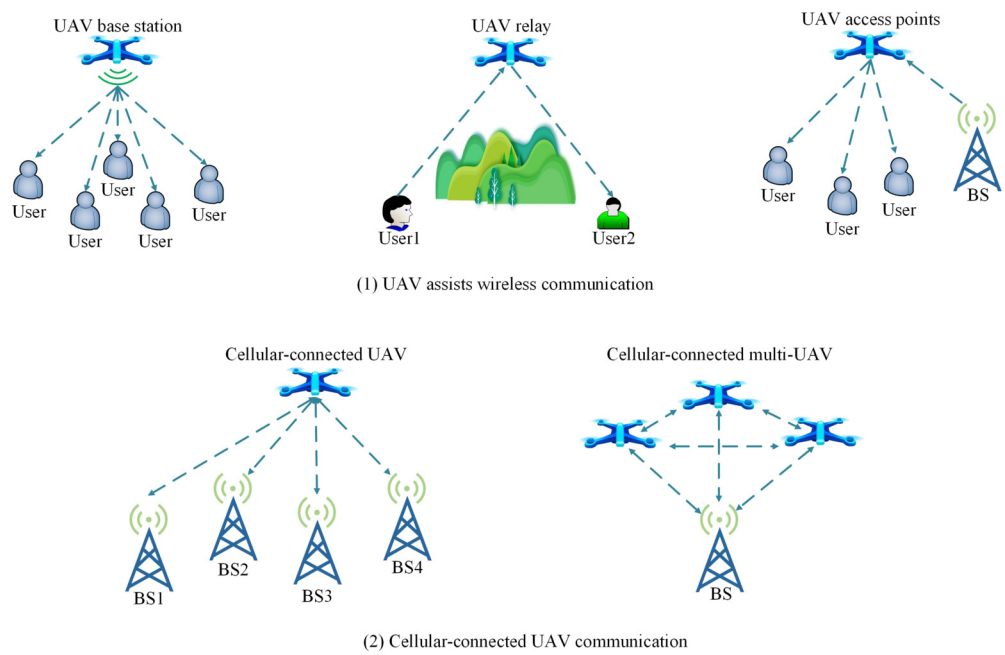

Fig. 2. The types of UAV communication.

$$E = E_F + E_C^C + E_C^D \tag{1}$$

Among these variables, E_F represents the propulsion energy consumption of UAVs, E_C^C represents the energy consumed for data calculation and processing, and E_C^D represents the energy consumed for the output of information.

We define the EE of the UAV as the ratio of the energy consumed by UAV in a certain time range and the number of bits consumed by information transmission [24], and it can be represented as the following Formula (2):

$$EE(q(t)) = \frac{\bar{R}(q(t))}{\bar{E}(q(t))} \tag{2}$$

$$R(q(t)) = \int_0^T B\log_2(1 + \frac{\gamma_0}{H^2 + \|q(t)\|^2})dt \tag{3}$$

Where $R(q(t))$ represents the functional relationship between the total amount of information R transmitted by the UAV and the flight trajectory $q(t)$ at time t, H denotes that the UAV flies horizontally at a fixed altitude H without frequent vertical ups and down, and γ_0 is the transmission signal-to-noise ratio (SNR).

3. Energy efficiency in trajectory planning and deployment

3.1. Propulsion energy consumption modeling of a fixed-wing UAV

This section focuses on reducing the propulsion energy consumption by optimizing the trajectory and three dimensional (3D) deployment, thereby improving the EE of UAVs. And before optimization, the propulsion energy consumption of UAVs should be modeled to clarify the relationship between each influencing factor and energy consumption.

According to [24], the propulsion energy consumption model of a fixed-wing UAV can be expressed by Equation (4).

$$\begin{aligned} E(q(t)) = \int_0^T \Bigg[& c_1\|v(t)\|^3 \\ & + \frac{c_2}{\|v(t)\|}\left(1 + \frac{\|a(t)\|^2 - \frac{(a^T(t)v(t)^2}{\|v(t)\|^2}}{g^2}\right)\Bigg]dt \\ & + \frac{1}{2}m\left(\|v\ (T)\|^2 - \|v\ (0)\|^2\right) \end{aligned} \tag{4}$$

The Equation (4) is explained as follows. The horizontal trajectory of the UAV in time period T is $q(t) = [x(T), y(T)]^T, 0 < t < T$, and the instantaneous velocity and acceleration can be expressed as Equation (5).

$$v(t) \triangleq \dot{q}(t) \qquad a(t) \triangleq \ddot{q}(t) \tag{5}$$

c_1 and c_2 are two parameters, c_1 represents the wing area and c_2 represents the air density, they both related to the weight of UAV. g represents the gravitational acceleration, and its standard value is 9.8 m/s^2. The letter m represents the total mass of the UAV and all its payloads. The first half of Equation (4) represents the energy required by the UAV to overcome the resistance and maintain the flying state, which is only related to the flight speed and acceleration of the UAV, and the latter half represents the change in kinetic energy generated after the UAV speed changes.

3.2. Propulsion energy consumption modeling of the rotary-wing UAV

The model in [25] only considers the influence of the flying speed of a rotary-wing UAV on the energy consumption. However, in many conditions, the UAV will change its speed and direction when flying, and these behaviors will no doubt lead to additional energy consumption. Therefore, it is very important to have a model that can more accurately represent the relationship between velocity, acceleration, direction change and energy consumption. In the latest research [26], a generalized propulsion energy consumption model (PECM) is proposed, which not only considers the impact of the speed on energy consumption but also considers the influence of acceleration and direction change on energy consumption. That model can be represented by Equation (6).

$$P(a,v) = P_i\kappa(a,v)\sqrt{(\kappa^2(a,v) + \frac{\|v\|^4}{4{v_0}^4})^{\frac{1}{2}} - \frac{\|v\|^2}{2{v_0}^2}} + P_0\phi(v) + \psi(v) \tag{6}$$

$$\phi(v) = (1 + \frac{3\|v\|^2}{U^2{}_{tip}}) \qquad \psi(v) = \frac{1}{2}d_0\rho sA\|v\|^3$$

Among these variables, P_0 represents the blade profile power and P_i represents induced power. $\psi(v)$ is the parasite power of the UAV. U_{tip} is the tip velocity of the rotor blade. v_0 represents the mean rotor induced velocity when hovering, d_0 is the fuselage drag ratio, s is the rotor solidity. ρ represents the density and A is the rotor disc area. Both velocity and acceleration can be expressed as functions of the trajectory $q[n]$, where

$$v[n] = \frac{q[n+1] - q[n]}{\delta_t} \qquad a[n] = \frac{q[n+2] + q[n] - 2q[n+1]}{{\delta_t}^2} \tag{7}$$

Then, Equation (6) can be represented as (8):

$$P[n] = P_0\phi[n] + \psi(v) + P_i\kappa[n]\sqrt{\left((\kappa[n])^2 + \frac{\|v[n]\|^4}{4{v_0}^4}\right)^{\frac{1}{2}} - \frac{\|v[n]\|^2}{2{v_0}^2}} \tag{8}$$

According to Equations (4) and (8), it can be seen that a uniform motion in a straight line saves the most energy because it does not need extra energy to change flight speed and acceleration. Reducing the occurrence of sharp turns and smoother trajectories can also decrease the energy consumption. When UAVs are hovering, the EE of UAVs is inefficient in terms of propulsion energy consumption, but for transmission channel conditions and throughput is the best. Therefore, for the optimization of the UAV trajectory, the best balance between the maximum system throughput and minimum EE must be achieved [24].

Combined with the energy consumption model, the following section reviews the trajectory optimization and 3D deployment.

3.3. UAV trajectory optimization

The traditional UAV trajectory planning problems can be abstracted into the traveling salesman problem (TSP) and pickup and delivery problem (PDP). Then, solving the problem can get the initial trajectory of the UAV. However, these models without considering the resource constraint problem of trajectory. Through the investigation and study of the relevant literature, the following rules for trajectory planning of the EE maximization problem can be concluded. In order to reduce the energy consumption of UAVs by optimizing the trajectory, an energy consumption model related to the speed, acceleration and other influencing factors of UAVs should be established firstly. Then, combined with the constraints of the related resources, optimization objectives should be proposed. However, such problems are typical of the nonconvex optimization problems and are difficult to solve, so they need to be transformed into convex optimization problems to obtain an approximate solution.

In this paper, we divide the solving algorithm of the convex optimization problem into the following categories: iterative algorithm, heuristic algorithm, reinforcement learning algorithm, recursive algorithm and phased optimization algorithm. Next, combined with relevant articles, the characteristics of the application of these algorithms in UAV trajectory optimization are as follows.

3.3.1. Iterative algorithm

The iterative algorithm is a method of continuously recursing new values of variables from the old values. In UAV trajectory planning, commonly used iterative algorithms include block coordinate descent (BCD), successive convex approximation (SCA) and simulated annealing (SA).

Authors in [27–29] used BCD to optimize the trajectory of UAVs. The idea of BCD is to optimize only one variable in the process of each iteration, while the remaining variables remain unchanged, and then solve them alternately. In the process of optimization, the original objective function is generally decomposed into two or more subproblems to be solved separately. For the solution of each subproblem, some parameters need to be fixed, and the solution needs to be iterated step by step until the optimal solution is found. Because the optimization process involves multiple parameters, the initialization of parameters is particularly important. As expressed in the above paper, when using the BCD to optimize the UAV trajectory for the purpose of maximizing EE, the optimal solution is obtained at the cost of throughput.

Authors in [30–33] optimize the trajectory based on the SCA, and authors in [25] and [34–36] make some improvements on traditional SCA. The idea of the SCA is to obtain new results from the existing initial conditions, and gradually approximate the original problem by tightening the conditions until the problem is finally solved. In trajectory planning, Taylor expansion of the objective function is often used to construct the approximate function, and then the approximate function is used to replace the original objective function to obtain the approximate solution. At present, SCA is the most widely used algorithm in the research of UAV trajectory optimization. According to the optimization results of [25] and [30], the throughput is not completely sacrificed in the results of this algorithm and can be considered in combination with throughput problems. It can be gleaned from [34] that the SCA has a higher computational complexity, but it is suitable for a system with a large amount of data.

BCD and SCA are the traditional methods often used in trajectory planning, and the use of these two methods usually requires that the specific parameters of the model are known and that accurate mathematical models can be established.

3.3.2. Heuristic algorithm

The heuristic algorithm is an algorithm that gives feasible solutions of each instance of the problem to be optimized and selects the best solution from them under acceptable time and space costs. At present, most heuristic algorithms are based on an imitation of natural algorithms, such as the ant colony optimization (ACO), SA, evolutionary algorithm (EA), etc. In addition, in terms of UAV trajectory planning, the covering path planning algorithm (CPP) and cuckoo search algorithm (CS) adopted in some papers are also heuristic algorithms.

Authors in [37] compare the heuristic algorithm with the exhaustive algorithm. In contrast, the latter one has a higher complexity while producing a better solution. But the heuristic algorithm can obtain a certain trade-off in complexity and quality of resolution.

ACO is proposed based on the principle of the ant foraging process. Authors in [38] proposed an improved algorithm called MACO, which overcomes the dilemma that heuristic algorithms are prone to fall into local optimal solutions.

SA is a heuristic random search algorithm based on a Monte Carlo iterative solution [39]. The idea source of it is to simulate the annealing process of solid material, and find the global optimal solution in the solution space by constantly generating new solutions and judging whether they are optimal. The advantage of this algorithm is that the solution obtained by the algorithm is independent of the initial solution state, while the disadvantage is

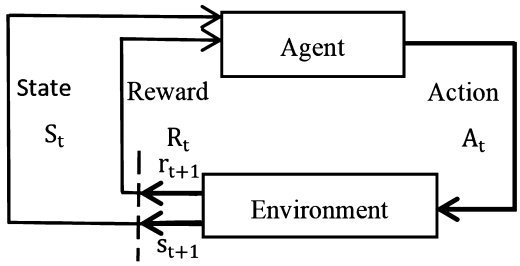

Fig. 3. Illustration of the reinforcement learning algorithm.

that the algorithm do not consider the maneuverability constraint of UAVs in most of trajectory planning, so the trajectory obtained by the algorithm is not very usable.

Differential evolutionary (DE) is a simple and effective EA, which has been successful applied in many fields. EA have been successfully applied to a very broad range of optimization problems with a fixed number of design variables [40], while DE was designed to be a stochastic direct search with unfixed variables for non-differentiable, nonlinear and multimodal functions based on it [41]. Due to the fact that the number of stop points of UAV is unknown a priori, the length of the individual is not fixed. DE is appropriate for energy-efficient trajectory planning for a Multi-UAV [42]. Among EAs, genetic algorithms (GAs) have been vigorous enough for various engineering optimization problems to find the optimal solution, which is also usually used for trajectory design in UAV-to-Ground communication [43].

CPP is an algorithm that covers a specific area containing obstacles with a minimal overlap and the shortest path. When applied to UAVs, the weight of the sensor, flight endurance, direction and the strength of the wind must be considered. In [44], decomposition methods of different types of regions are summarized. Authors in [45] proposed a new coverage path planning algorithm to reduce energy consumption through minimizing the number of rotations covering the region.

CS is an algorithm that can effectively solve optimization problems by simulating parasitic brooding of cuckoo species. Relevant studies show that CS is better than the other colony algorithms, such as the particle swarm optimization (PSO) and ACO. This algorithm is used in [46], which not only optimizes EE but also effectively improves the key indicators such as system throughput and task delay.

3.3.3. Reinforcement learning

Reinforcement learning refers to a machine learning algorithm that uses new data to repeatedly update and iterate the existing model. Its characteristic is that in the optimization process, it can not only use the existing data but also obtain new data through the exploration of the environment to achieve the purpose of real-time dynamic optimization. Deep reinforcement learning (DRL) refers to all reinforcement learning algorithms that use neural networks as parameter structures for optimization.

As shown in Fig. 3, in the reinforcement learning algorithm, the purpose of the agent is to obtain the action with the maximum reward value in the interaction time. By sensing the state of the environment, the agent reacts according to the current state and inputs the action to the environment. The environment outputs the next state and the reward obtained by the current action according to the action taken by the agent.

Authors in [47–49] adopted the idea of deep reinforcement learning to achieve the purpose of real-time dynamic path planning. Authors in [48], based on the idea of DRL, proposed energy-efficient fair communication through trajectory design and band allocation (EFC-TDBA), which overcomes the problem of user mobility that traditional algorithms cannot solve. In the meantime, EFC-TDBA reduces the complexity of the algorithm and improves the accuracy.

3.3.4. Recursion algorithm

The recursion algorithm is a kind of method to solve problems by repeatedly decomposing problems into similar subproblems.

Dynamic programming (DP) is one of the recursion algorithm, which decompose the original problem into a subproblem that is easy to solve, and these subproblems often do not exist independently. The solution of each subproblem is solved first, and then the optimal solution of the original problem is obtained from the solutions of the subproblems. DP improves a search algorithm with exponential time complexity into one with polynomial time complexity, which significantly reduces the algorithm complexity.

The conclusion in [50] also confirms that the complexity of DP is significantly reduced compared with heuristic algorithms and exhaustive algorithms, and its performance is not much different from that of exhaustive algorithms.

3.3.5. Phased algorithm

The idea of this algorithm is to decompose the complex and difficult target into multiple stages to solve the problem. First, the influencing factors of the complex problem should be determined, then the solutions of the influencing factors are determined by stages, and finally, the optimal solution of the objective function is obtained step by step.

The energy-efficient trajectory planning algorithm (TPA) proposed in [42] is a phased algorithm. In the first stage, the differential evolution algorithm with a variable population size is used to update the position and number of users. In the second stage, the k-means clustering algorithm is used to cluster the users. In the third stage, a low-complexity greedy method is used to construct the order of the users in each cluster to quickly generate the trajectory of the UAV, and the simulation results also show that this algorithm has a significant improvement in EE. In [23] and [51], the optimization objective is also divided into two stages, the final optimal solution can be obtained by an alternate optimization of the problems in the two stages, and the overall complexity can be reduced.

Table 4 summarizes and compares the above algorithms, so that readers can have a clearer understanding of the characteristics and the use of the above algorithms in UAV trajectory optimization.

By summarizing the typical UAV trajectory optimization algorithms mentioned above, it can be seen that each algorithm has its own characteristics, advantages and limitations. Therefore, appropriate algorithm can be chosen with considering the target problem and expected result for the optimization problem. For example, an iterative algorithm can be considered when the specific parameters of the model are known and accurate mathematical models can be established. The heuristic algorithm can be considered when obtaining the trade-off between complexity and quality. To obtain the real-time trajectory of the UAV, reinforcement learning algorithm can be considered. If the problem is complex and there are many influencing factors, the phased algorithm is a good choice. In addition, it may be considered to combine or improve the use of various algorithms to draw on each other's strengths to obtain better results. For example, this point of view is reflected in [47] and [52].

3.4. 3D optimized deployment of UAVs

In the previous subsection, the trajectory planning of the horizontal position of the UAV was introduced. However, in reality, the UAV flies in 3D space, so the energy consumption is also affected by the altitude of the UAV. The transmission distance be-

Table 4
Comparison of energy efficient algorithms in trajectory planning.

Algorithm		Article	Main idea	Features	Performance	Advantages	Limitations
Iterative algorithm	BCD	[27–29]	Different variables are solved alternately	At the expense of throughput	Convergence rate is fast	Simple and efficient	The code is less concise than a recursive algorithm
	SCA	[30–33]	Gradual approximation	Suitable for a system with a large amount of data	High computational complexity	The non-convex problem can be transformed to a convex problem	There is no guarantee that the resulting solution is globally optimal
Heuristic algorithm	MACO	[38]	The metropolis criterion is introduced on the basis of ACO	Avoid local optimal solution states	Lower complexity than ACO	Overcoming the problem that ACO is easy to fall into the local optimum too early	The optimized trajectories may be discontinuous
	SA	[39]	New solutions are constantly generated to judge whether they are optimal or not	The optimized trajectory is not very usable	Slow convergence but high accuracy	The solution is independent of the initial solution state	The maneuverability constraints of the UAV are not considered
	EA	[40–43]	Survival of the fittest	Appropriate for energy-efficient trajectory planning for a multi-UAV	Converges faster	The structure of the algorithm is simple and it is easy to implement	May be trapped in local optimal point
	CPP	[44,45]	Cover every point of a certain area of interest	The weight of the sensor, flight endurance, direction and the strength of the wind must be considered	Less computational time	High coverage rate	It is difficult to apply to complex scenarios
	CS	[46]	Search a better solution to replace the bad one	Overall optimization effect is better	Computing efficiency has been optimized	The operation of the algorithm is simple and easy to implement	It is not suitable for problems with large solution space
Reinforcement learning	DRL	[47–49]	Update the existing model repeatedly with new data	The problem of user mobility is solved	Low time complexity than random and greedy search	Applications are extensive	Learning efficiency may not be high situation
Recursion algorithm	DP	[50]	The original problem is decomposed into similar sub-problems to solve	Can achieve almost the same outage performance as the exhaustive search	The algorithm complexity is lower than heuristic algorithms	Clear structure and highly readable	Taking up an additional space overhead
Phased algorithm	TPA	[42]	The complex problem is divided into multiple stages to solve	Optimization result is better than a single algorithm	The algorithm complexity is lower	Algorithms can be used together	The operation is more complex

tween ground users and the UAV increases with the altitude, which also leads to the increase of path loss in the transmission process. Because the path loss is generally caused by the transmission environment or channel conditions. Farther the transmission distance is, the greater the loss is.

In addition, another reason for 3D deployment of UAVs is that the operation ability of one UAV is limited, and multiple UAVs are often required to cooperate to complete the task. At this time, coverage between multiple UAVs and the area of the ground user should be considered to achieve optimal coverage to reduce the transmission energy consumption.

In downlink scenes, authors in [53–55] considered the effect of UAV altitude on energy consumption. The conclusion in [55] indicates that the total transmitted power of the downlink is markedly reduced by making the UAV achieve the best coverage to determine the deployment of the UAV. Then, the transmission energy consumption of the UAV can be minimized through adjusting the altitude of UAV.

Authors in [53] analyzed the effect of the altitude of UAV on the overall performance, and given the method to find the optimal flying altitude of the UAV. In [54], the flying altitude and the number of UAVs are also considered. And by coordinating the flying altitude and the number of UAVs, the energy consumption of a single UAV can be minimized.

Optimal coverage and energy consumption of multiple UAVs were studied in [56] and [57]. Authors in [56] proposed a coverage model of multiple UAVs based on energy-saving communication. Different from traditional methods, this paper adopted the concept of coverage probability to describe the coverage situation, making the coverage model more realistic and effective. In addition to reducing the transmission power of UAVs, [57] also satisfies the minimum ground user rate and system load balance conditions.

In [58], the influence of the optimal coverage and altitude of the UAVs on transmission power is considered jointly, and the functional relationship between the coverage probability of the UAVs and the altitude and antenna gain is given, which provides a theoretical tool for subsequent related research. In [59] and [60], the UAV altitude and path loss factor are jointly optimized. In [59], low energy consumption is achieved through joint optimization, and at the same time, better coverage and throughput performance is achieved. Several heuristic algorithms that can be used in joint optimization are compared in [60], including genetic algorithms, particle swarm optimization and non-hierarchical methods. Their performance is tested, showing the advantages and disadvantages of the algorithm.

Through a study of the literature, it can be known that the flying altitude of UAVs is not invariable, and the altitude between

the group of UAVs is also different. Therefore, when studying the effect of height on energy consumption, simply fixing the height does not produce the most satisfactory solution. Multiple factors should be jointly optimized, and by considering the influence of dynamic changes in height on energy consumption, the scheme will be more practical.

4. Energy efficiency in resource allocation and management

With the continuous development of 5G technology, the number of mobile users is also experiencing an explosive growth. According to the data in the 2020 Global Network Trend Report, the number of connected devices in the world is expected to reach 48.9 billion by 2023, resulting in greater resource demands. UAVs are characterized by rapid deployment, high flexibility and low cost. Therefore, in high-density communication user scenarios, UAVs are often used as temporary BSs or relays to assist wireless communication [61]. Compared with low-density scenarios, high-density scenarios require more computation and data transmission to UAVs. This undoubtedly increases the total energy consumption of the UAV communication system and reduces the life cycle of the whole system because of the limited endurance.

In the previous section, the components of energy consumption in UAV communication were analyzed, and the propulsion energy consumption was modeled and optimized. In this section, we will optimize the communication energy consumption of UAVs. According to the previous introduction, the communication energy consumption of UAVs can be expressed as

$$E_C = E_C^C + E_C^D \tag{9}$$

For transmission energy consumption E_C^D, assuming that the UAV is a single-antenna device, the total bandwidth and the time duration of the system are divided equally into N_F subcarriers and N time slots, respectively [62], so

$$E_C^D = \sum_{K=1}^{K} \sum_{i=1}^{N_F} a_k^i[n] p_k^i[n] \tag{10}$$

Where $p_k^i[n]$ represents the power allocation of user K on subcarrier $i \in \{1..., N_F\}$ at time slot n. Variable $a_k^i[n] = 1$ represents that subcarrier i is assigned to user k at time slot n; otherwise, $a_k^i[n] = 0$. For E_C^C, it is a function of the computational frequency of the CPU.

In conclusion, the UAV communication system can save energy consumption and improve EE if the UAV computing resources or network resources such as channel, bandwidth and power can be reasonably coordinated and allocated.

4.1. Network resource allocation

4.1.1. Bandwidth and channel allocation

Due to the characteristics of electromagnetic waves, the spectrum that can be used for communication is limited. With the development of 5G networks, the emergence of mass mobile users makes spectrum resources increasingly scarce. To better meet the demand for spectrum, long-term evolution (LTE), non-orthogonal multiple access (NOMA), multiple-input-multiple-output (MIMO) and other technologies are considered to be good methods for the effective utilization of spectrum resources and can combine bandwidth and channel allocations to produce better EE optimization results [63].

The Third Generation Partnership Project (3GPP) proposed licensed-assisted access (LAA) and applied LET to the unlicensed frequency band to expand the available bandwidth. Based on the frequency sharing mode, bandwidth allocation problem were discussed in [64] and [65]. Authors in [64] adopted orthogonal frequency division multiple access (OFDMA) to make the orthogonal use of bandwidth between multiple UAVs; thus, bandwidth utilization and EE can be improved. In addition, [65] is a scenario of UAV data collection.

The subchannel using NOMA technology can be shared by multiple users at the same time, which greatly increases the channel capacity of the system and improves the transmission efficiency. Authors in [62,63] and [66,67] studied the subchannel allocation problem, and there are two main approaches to the research methods in these articles. One is to classify users according to different conditions or constraints and then to match the channel with the lowest EE target for the whole user group [62], [66]. Another is to match each user in turn. In [63], the subchannel allocation problem is modeled as a process of bilateral match-swapping. First, each user will initially match the subchannel based on the principle of maximizing the data rate. Second, the previously matched users are swapped in pairs for the purpose of maximizing EE so that one user can be optimized in each exchange. In [67], channel matching is directly implemented for users with the goal of using minimum transmitting power and transmitting to one user at a time successively. Therefore, the algorithm complexity of the first idea is lower than that of the second algorithm. Of course, channel allocation should not only pursue EE maximization but also exert a large influence on the other system indicators. Therefore, authors in [62] and [66] take into account the satisfaction of quality of service (Qos). In addition, the system throughput and delay time should also be important indicators to be considered.

4.1.2. Power allocation

The previous subsection described ways to increase EE through the allocation of bandwidth and channel, and the next subsection will describe the ways to reduce energy consumption by coordinating the power allocation. In the existing papers, the power allocation for UAVs is mostly optimized jointly with the trajectory planning and deployment of UAVs to minimize the transmission power of UAVs and improve the power utilization rate.

NOMA is also widely used in power allocation. NOMA technology adopts power multiplexing technology, at the transmitting end, UAVs allocate power to the users according to relevant algorithms. In [68], it is also proved that, compared with orthogonal multiple access technology (OFM), the EE of NOMA technology is improved. This technique is applied in the literature [69–71]. In [69], power allocation is carried out by a Lagrange dual analysis. [70] uses an iterative algorithm, which transforms non-convex problems into convex problems through an approximate transformation.

Except for NOMA technology, spectrum sensing is used to control whether data are transmitted or not in [72] to achieve the purpose of controlling the power allocation. As the working environment of UAVs is dynamically changing, resource allocation schemes that can adapt to the dynamic environment are urgently needed. Authors in [73] introduced the embedded UAV communication system to realize the real-time optimization of UAV resource allocations. [74] combines the 3D deployment and power allocation of UAVs and simultaneously optimizes the transmission power of UAVs from two perspectives, which is a case worth referencing in our future research.

Research for network resources allocation was summarized in Table 5. From this table, we found that the main idea of resource allocation method is to try to improve the utilization rate of resources in short supply, and reasonable resource allocation can bring about the improvement of EE. In the energy consumption composition of UAVs, the proportion of propulsion energy consumption is much greater than that of communication energy consumption. However, some authors do not consider the propulsion

Table 5
Comparison of energy efficient methods in resource allocation.

Methods	Article	Main idea	Features	Performance	Advantages	Limitations
Bandwidth allocation	[64]	Improved EE by increasing bandwidth utilization	Frequency sharing	Improved EE while satisfying the QoE	The algorithm extends to multiple UAV networks	The issue of interference between multiple UAV is not addressed
	[65]			The performance is more superior to that of the greedy based algorithm	Strong robustness	The study is limited to the scene of a single cellular user-UAV
Channel allocation	[62]	Improved EE by increasing channel capacity	User clustering	Significantly improved EE while meeting the constraints of communication security	The dynamic trajectory of the UAV was designed	Failure to consider the overall EE in combination with flight trajectory
	[66]			Maximized the throughput by achieving the optimal coverage	Deploy UAVs accurately and quickly	The dynamics of UAVs are not considered
	[63]		A single user matched	Improved EE while applying caching technology to reduce latency	The power allocation is also optimized	The dynamics of UAVs are not considered
	[67]			Improved EE and achieved load balance	Considered the constraints on QoS, network capacity and power consumption	Only obtain an approximate solution due to the increased computational complexity
Power allocation	[69]	Improved EE by power reuse	NOMA	Maximized total energy efficiency with quality of service	Jointly considering the user scheduling, transmit power allocation and UAV trajectory optimization	The UAV altitude changes were ignored
	[70]			Improved the EE while meeting the requirements of QoS	Research based on mm Wave-enabled NOMA-UAV networks	The dynamic variability of the UAV was not considered
	[71]			Jointly maximize the EE and the downlink throughput fairness among users	Low computational complexity	Research was based on a single UAV
	[72]	Improved EE by controlling data transfer	Spectrum sensing	Achieved the optimal assignments on sensing time and secondary transmission power	Significantly better than PSO algorithm on improving the EE of UAV-based cognitive radio system	UAV flies on a fixed trajectory

energy consumption in their optimization, and only optimize the communication energy consumption, which is limited to change the overall EE. Therefore, this is a very worthwhile question in future research to make our study more valuable.

4.2. Calculating resource allocation

In the current research on computing resource allocation, the authors mainly apply the method of MEC. As early as 2014, The European Telecommunication Standards Institution (ETSI) first proposed the MEC concept [75]. Its ultimate goal is to provide intelligent services and some cloud computing functions for mobile users and to create a carrier-level service environment with high performance, low latency and high bandwidth based on the principle of approaching wireless access networks [76].

Compared with cloud computing technology, MEC shortens the distance between cloud devices and edge users and reduces communication energy consumption and delay. However, new networks and new mobile applications, such as mobile games and speech recognition, put forward higher requirements on network quality and require more computing resources to meet its low delay and high reliability communication [61]. Because the traditional MEC server is fixed in location, it cannot be flexibly changed according to the movement of the mobile users, which limits the ability of the MEC server. Sometimes it is even difficult to establish the ground MEC network due to the harsh environment. Therefore, using UAVs to deploy MEC will play a very important role. MEC has become a key technology in the development of 5G networks. According to the introduction in [77], there are three models for deploying MEC using UAVs, as shown in Fig. 4.

Next, the MEC computing resource allocation problem in the three models will be described.

4.2.1. UAV serving as a computing server

When the ground MEC network cannot be reliably established, the UAV can act as an MEC server to help ground mobile users perform computing tasks.

In UAV-assisted MEC technology, the main application model is to use UAVs to carry MEC servers and provide unloading services for users on the ground. This approach is suitable for a variety of scenarios, such as device-to-device communication (D2D), IoT, UAV-assisted cellular networks, UAVs as BSs, communication in hot spots, and communication in areas where network infrastructure has been severely damaged. However, the computing resources and energy of UAVs are limited, so these resources must be rationally allocated to improve the EE of UAVs. The author of [78] considers the allocation of MEC computing resources entirely from the perspective of profit and cost, without taking into account the lim-

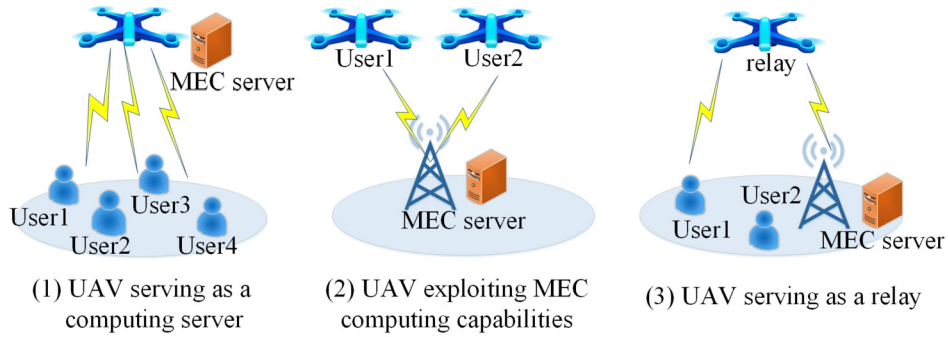

Fig. 4. Three models of MEC.

itation of UAV energy. Both [61] and [79] take the maximization of UAV EE as the goal in studying the MEC computing resource allocation problem. The former obtains an approximate solution through the greedy algorithm, while the latter uses ant colony system algorithm to solve for the optimal solution. Then, in [80], the model was extended to multi-UAV scenarios.

However, a good UAV communication system should not only consider the minimization of energy consumption in the calculation process but also meet constraints such as user QoS, computing delay time, system security and minimization of UAV flight energy consumption.

Network latency. The main point of using UAVs as MEC servers is to improve the data calculation efficiency and reduce the delay. When optimizing the energy consumption of UAVs through resource allocation, meeting the minimum delay time for each task is also a key index. Minimizing the energy consumption and mission completion time of the UAV are taken as the objectives of the resource allocation process in [81]. The Pareto optimal solution, balancing UAV energy and completion time, is also studied, and the delay is controlled by controlling the completion time. In [82], the author firmly took the minimization of average delay as the optimization objective, and the energy constraint was satisfied in the optimization process. In contrast, the practice of [83] better meets the requirements. Based on IoT application scenarios, the maximum allowable delay time of each task and the energy budget of the UAV are taken as constraint conditions. This paper optimizes the uninstall strategy through an alternative optimization algorithm to maximize the EE. However, it only considers the single UAV model, and the unloading and computing capacity are limited. Therefore, authors in [84] used D2D communication equipment as an additional computational unloading option to further reduce the energy consumption and time delay of UAVs. In [85] and [86], the system integrity is taken into consideration, and the energy consumption of user equipment (UE) and UAVs is taken as the optimization target, effectively extending the life cycle of the whole communication system.

Trajectory of flight. On the one hand, the trajectory of UAVs directly affects the propulsion energy consumption of UAVs. On the other hand, to reduce the transmission distance and energy consumption, its position change also indirectly affects the user's unloading strategy to the UAV. Therefore, it is necessary to jointly optimize the trajectory of the UAV and the resource allocation to obtain greater EE gain. In [87], under the constraint of flight speed, the UAV uses a circular trajectory flight to unload data from the BS and adjusts the trajectory of the flight and the resource allocation jointly by BCD to maximize EE. At the same time, the authors of [88] and [34] used the SCA to solve the problem. The author of [34] compared the SCA algorithm with decomposition and iteration (DAI)-based algorithm, thinking that the SCA algorithm should be used to find the optimal solution for a relatively large amount of data. The DAI-based algorithm is a better choice to attain a smaller computing energy consumption for a relatively small amount of data.

Authors in [89] extended the optimization objective from the energy consumption of a single UAV to minimize the weighted energy consumption of UAVs and UEs. In [90], the 3D trajectory was designed for the first time in a UAV-assisted MEC system when jointly optimizing the UAV trajectory and resource allocation energy consumption, which is more valuable in application.

System security. The communication between ground users and UAVs is easily intercepted by nearby eavesdroppers [91], resulting in information leakage. On the one hand, information security issues will arise. On the other hand, information retransmission may be a reality, resulting in resource waste. Therefore, it is necessary to consider the safe unloading problem of the MEC system while minimizing energy consumption in the process of resource allocation. Authors in [92] used the method of UAV self-jamming to improve safety and showed that there is a fundamental trade-off between the safety and delay of UAV MEC systems. In [93], the difference of convex algorithm (DCA) was used to obtain the minimum energy consumption of the MEC system under the constraints of the confidential transmission rate and the computational delay. In [94], a system function with task delay and computing capability (i.e., the weighted sum of the energy consumption and eavesdropping rate) was obtained by jointly optimizing the unloading rate and resource allocation.

4.2.2. UAV exploiting MEC computing capabilities

When the UAV has limited computing abilities and needs to perform computation-intensive tasks, it can offload the tasks to the ground MEC server for computing as a user. In this model, the UAV directly reduces its own computing energy consumption greatly by using MEC computing resources. In the unloading process, the amount of energy consumed depends on the unloading transmission power and unloading time.

Authors in [95] studied random calculation resource management based on the Lyapunov optimization algorithm and obtained the optimal transmission power and bandwidth allocation of calculation unloading alternately to maximize the system utility. Authors in [96] extended the model to a multi-UAV user system, and the energy consumption of all UAVs was minimized by a resource allocation strategy on the premise of satisfying all user delay constraints. Similarly, UAVs also have a risk of information leakage when unloading tasks to the MEC server. Therefore, security issues are considered in [97] and [98]. Authors in [97] establish a security model including eavesdroppers to minimize UAV energy consumption under time and security constraints. A highly decentralized chaotic replacement box (S-box) is used in [98] to encrypt each computation task. The problem is described as an energy consumption minimization problem, and two heuristic algorithms are proposed to solve it.

4.2.3. UAV serving as a relay

When the offloading link between the mobile user and the ground BS (with MEC server) is poor, the UAV acts as a relay to assist the mobile user in the offloading tasks to the ground BS.

To date, there are few studies in this direction. In [99], a mission prediction algorithm based on long short-term memory (LSTM) is used to dynamically predict the task load unloaded to the UAVs, enabling the UAVs to arrange calculation and unloading strategies in advance according to their own computing and storage resources to obtain the best EE solution. In addition to optimizing the resource allocation of UAVs, the trajectory of UAVs is also jointly optimized to maximize the EE of the UAVs in [100].

Because UAVs are easy to deploy and highly flexible, a growing number of studies are using UAVs to deploy MECs. However, the articles fully considered that the QoS computational delay, system security and trajectory is lacking, especially in the model of UAVs exploiting MEC computing capabilities, and the UAVs serving as relays, so there is considerable room for research exploration in the future.

5. Energy efficiency in communication protocol design

UAV communication has been widely used in all walks of life. However, at present, most applications remain at the level of single UAV operations, which limits the operating area, carrying payload and working efficiency. A form of multi-UAV cooperation can effectively solve these problems and has begun to be applied in a variety of scenarios. To carry out appropriate cooperation among multiple UAVs, it is necessary to form a UAV wireless communication network or UAV ad hoc network among UAVs [101], which does not depend on communication infrastructure and can be connected quickly. The characteristics of the UAV ad hoc network are that it is dynamical and is limited by the energy of the UAV. Therefore, the traditional network communication protocol is not fully applicable to the UAV ad hoc networks. In this respect, we can design an appropriate communication protocol to meet the demand of dynamic change and to improve EE to prolong the network life [102]. Next, we will review the communication protocols of the network layer and data link layer.

5.1. Network layer

The network layer is mainly used to implement the transparent transmission of data between two end systems so that the data of the source end can reach the destination end through multiple nodes in the subnet with the best path, which is the transmission routing problem. We usually divide routing protocols into static routing and dynamic routing. Due to the complex network structure and high mobility of UAVs, it is a challenge to design efficient dynamic routing protocols to support such networks [103]. The dynamic routing protocol is a multihop routing protocol. How to select nodes for each hop or how to find the best next-hop route to adapt to dynamic topology changes, reduce energy consumption and prolong network life is the main challenge we are facing [103].

There are four classical routing protocols for UAV ad hoc networks [104]: ad hoc on demand distance vector (AODV), dynamic source routing (DSR), optimized link state routing (OLSR), and geographic routing protocol (GRP). Through the simulation results, it is found that AODV has a lower data discarding rate, DSR has higher received traffic, and OLSR has lower network delay and higher system throughput. However, these protocols do not consider EE and achieving the best performance is not among their advantages. In recent years, on the basis of the above classical protocols, the authors proposed an improved routing protocol for the improvement of EE. Here, we divide these protocols into the following three types according to their different multihop node selection modes: routing protocols based on path selection; routing protocols based on node selection; and routing protocols based on clusters [105,106].

5.1.1. Routing protocols based on path selection

In this kind of method, when choosing the intermediate nodes, we mainly consider the transmission distance and ensure that the energy consumption of this path is minimal.

A swarm energy-efficient multicast routing protocol (SEMRP): This protocol is a multicast routing protocol based on the principle of minimum distance to build a multicast tree when the source has data to send to multiple destinations. This method is helpful to reduce the number of hops in the process of data transmission, thus reducing energy consumption, and the simulation results also prove that the SEMRP protocol achieves good results in the transmission of multicast messages in UAV groups [107].

An ad hoc routing protocol for FANETs based on greedy forwarding and limited flooding (PSO-GLFR): This protocol is optimized by the greedy forwarding and flooding-based routing protocol (GFR) by applying PSO. The GFR preferentially adopts the greedy forwarding policy, sending the data to the neighbor node nearest to the destination first. When this policy fails, the method of flood routing is selected to find the path but this method only considers the distance problem, and the result may not be the optimal path. Therefore, authors in [108] use PSO with energy consumption minimization as the objective function to optimize the protocol by considering constraints such as distance, transmission hop number and the number of neighbor nodes. Simulation results show that the optimized algorithm improves both delay time and energy consumption.

Energy-efficient multihop data routing (EMDR): This protocol can be used to improve EE and latency. Authors in [109] studied the EMDR algorithm, which can meet the quality of service and minimize energy consumption. The greedy algorithm is first used to select a path with the lowest energy to send data to the BS. If the UAV on the path is almost out of power at this time, it can abandon this path and choose another suboptimal path.

The multiobjective routing protocol (MORP): This protocol is based on the minimization of the multiobjective cost function to select the transmission path. The cost function reflects the different energy consumed by the UAV when transmitting information of size M and the residual energy of the UAV. The fairness of the energy consumption of different UAVs is ensured through this protocol, thus extending the network lifetime [110].

5.1.2. Routing protocol based on node selection

In this method, energy thresholds is set for all the UAV nodes and exclude nodes lower than the energy threshold when selecting the next hop node to maintain the durable running of the system.

Multiobjective optimized link-state routing protocol (MO-OLSR): This protocol developed based on the OLSR protocol, which is one of the most commonly used routing protocols in ad hoc networks. The energy threshold is set to help each improved algorithm select multipoint relay nodes, and the network delay, load and link stability are considered at the same time. When determining the optimal route, the nodes that meet the energy threshold are screened from the neighbor nodes first, and then the buffer check is performed on the nodes that meet the energy threshold to exclude the overloaded nodes to reduce the network delay. Finally, the stability of each link is checked [103].

Energy and direction aware location aided routing protocol (ED-LAR): This protocol considers the remaining energy and moving direction of the UAV when selecting the next hop node and defines the utility function (UF). The node with the lowest UF value among the neighboring nodes was taken as the next hop node, of which $UF = \alpha(1-\eta) + (1-\alpha)\theta$, η is the residue energy ratio, and

θ is the moving direction of the UAV, which indicates the angle of moving UAVs from the baseline draw from the forwarding UAV to the destination UAV, α is an adjustable weight of η and θ [111].

5.1.3. Routing protocols based on clusters

Routing protocols based on clusters have attracted extensive attention due to the extensibility, reliability, low latency and other advantages of clustering [112]. In this type of algorithm, we first divided the UAV nodes into several clusters according to the location of node mobility, and each UAV node in the group was assigned different tasks. The cluster head (CH) is used to perform the communication between the cluster or the data forwarding. A cluster gateway (CGW) is used to access the cluster's neighbors and exchange data with neighbors. The other nodes are cluster members (CMs) [106]. This protocol is easy to expand and can adapt well to the fast mobility and dynamic topology of the UAV networks.

UAV-assisted data gathering (UADG): This method is a cluster-based protocol. When selecting CH, the UAV with the most remaining energy in the cluster is selected, and other nodes without data transmissions are placed in the sleep state to save energy consumption [113].

A whale optimization algorithm based on optimized link state routing (WOA-OLSR): This method is also a cluster-based protocol. When selecting the next hop node, the number of neighbor nodes of each UAV is evaluated first, and the UAV with the largest neighbor nodes is selected for maximum packet transmission to reduce energy consumption [114]. The simulation results show that WOA-OLSR is superior to OLSR in throughput, delay time and energy utilization.

Low-energy adaptive clustering hierarchy (LEACH): This method is also a cluster-based routing protocol, but its CH selection is determined based on the size of the random number, which can make the low-energy nodes become CH to reduce the service life of the network. This protocol has been optimized in [115]. In the process of CH selection, the energy of each node and the residual energy information must be obtained. The node meeting the energy expression can be selected as the CH, and the other nodes are composed of clusters with the nearest CH. In the absence of a task, CM can enter the sleep state to minimize the energy consumption.

Energy aware link-based clustering (EALC): This protocol uses the k-means sorting fitness algorithm to group nodes into clusters. If the fitness of the CH is lower than a certain threshold, the cluster will be disbanded, and all nodes will become non-cluster head nodes. Then, the nodes should be reclustered according to the k-means algorithm. The simulation shows that this protocol is superior to the clustering model based on ACO [116].

Cluster-head restricted energy efficient protocol (CREEP): In this protocol, the initial energy of the nodes is not balanced, and the senior node energy is less than that of the ordinary nodes. The probability of the different nodes becoming CHs is designed according to the distance between the different nodes and BSs. According to the simulation results, by selecting the top 10% of the nodes as the CHs, the UAV ad hoc network can obtain the highest survival period but the system throughput will be affected [117].

5.2. Data link layer

A data link is on a physical link that controls the transmission of data through certain procedures or protocols to ensure the data is transmitted correctly. It is between the network layer and the physical layer, providing services to the network layer based on the services provided by the physical layer, and defining how data is transmitted over a single link. In a broadcast link, data transmission may cause interference or conflict when two or more nodes transmit at the same time, resulting in transmission failure. Multiple access control protocol (MAC) uses distributed algorithms to determine how nodes share channels, in other words, it decides when nodes can transmit data. The authors of [118] conducted a detailed investigation of previous MAC protocols and presented the challenges of designing MAC protocols for UAV communication networks, such as end-to-end latency, network capacity, EE, channel access control and secure transmission. The detailed implementation steps of eleven kinds of MAC protocols are given. The problems they solved are also analyzed. Among them, only energy-efficient and location-aware medium access control (EL-MAC), a kind of energy saving and location-aware medium access control protocol, achieves the goal of Qos through collision avoidance, congestion control, energy perception, etc. [119]. In [120], the author proposed an energy-efficient and fast MAC (EF-MAC) protocol based on wireless sensor networks, clustering sensor nodes, dividing time into time slots and prioritizing according to the remaining energy of the CH. The UAV receives data from the CH separately through different time slots, which greatly reduces retransmission due to collisions or conflicts, so it can reduce energy consumption while also outperforming traditional MAC protocols in terms of network latency. Therefore, research considering energy efficiency in MAC protocol design is still relatively lacking.

We summarize the literature on energy-saving routing protocols for the network layer and the data link layer in Table 6, where characteristics and advantages of each protocol were presented clearly. It is obvious that these protocols that give good results at a certain layer, but all of these literatures except [121] may not enhance overall energy maximization across all layers of the network. That is the biggest disadvantage of these literatures. They only solve the problem of energy loss of one layer while ignoring the consideration of energy consumption of the other layers. Authors in [121] proposed a cross-layer protocol called enhanced energy conservation based on residual energy and distance (EE-CRED), which took into account the energy consumption at the network layer and the data link layer. Compared with the performance of the single-layer protocol, the simulation results show that the cross-layer protocol has a very high EE and significantly improves the network lifetime and stability. Therefore, it can be seen that the cross-layer communication protocol is of great help in improving EE. However, relevant studies are not abundant at present, which offers a feasible research or improvement direction for our subsequent work on EE improvement.

6. Energy efficiency in energy harvesting and transfer

In recent studies, more and more articles are devoted to studying the energy supplement method of the UAV communication process. Although this method will not directly reduce the energy consumed by the UAV communication system, the method introduced by those studies can offer a constant source of power for the UAVs and prolong the life cycle of the UAV communication system. Combining this technology with the methods introduced above will be able to solve the EE problem in UAV communication. According to the description in [122], there are three main methods for energy supplementation in UAV communication at present (as illustrated in Fig. 5).

(1) Multiple UAVs work in circulation, replacing the UAV running out of battery with a new fully charged UAV;
(2) Replace the power depleted battery with a new one;
(3) Energy collectors are used to collect energy.

Table 6
Comparison of energy efficient protocols in UAV communication.

Protocols		Article	Main idea	Features	Performance	Advantages	Limitations
Network Layer	SEMRP	[107]	A multicast tree is constructed based on distance minimum principle	Real-time, Used for UAVs flying in group formations	Improved EE by reducing the total emission energy	End-to-end latency and packet delivery rates have been improved	Not universal
	PSO-GLFR	[108]	Shortest propagation path	The UAV node's own factors are considered comprehensively	Can provide low-latency, low-energy routing support for UAV-FANETs	Reduced delay time	Computational complexity is high
	EMDR	[109]	Energy minimum path	Suitable for disaster relief scenarios	The total required power to transmit is significantly decreased, and the network life cycle is increased substantially	Dramatically reduced computational complexity	Needs to be extended to many application scenarios
	MORP	[110]	Select the propagation path based on how much energy the UAV has left	Mobility and energy-aware	Minimize a weighted average of the different energies consumed by the UAVs	Achieving the fairness in energy consumption of UAVs	Flight area and altitude were restricted
	MO-OLSR	[103]	Set the energy threshold	Suitable for high-density UAV network	High energy performance	High network latency and system stability	Three or more hop neighbors will increase the system complexity
	ED-LAR	[111]	The remaining energy and direction of movement of the UAV	Energy and direction aware	Improved the link duration time between the UAVs	The link duration is extended	Other performance indicators were not considered
	UADG	[113]	Clusters according to the residual energy	Suitable for any density network	Gains the performance efficiency in terms of both execution time and energy consumption of the network	High time efficiency	Limited communication range
	WOA-OLSR	[114]	Clustering	High transmission security	Energy utilization and time complexity have been improved	End to end delay and throughput have been improved	Time complexity is high
	LEACH	[115]	The selection of CH is grounded on the new optimized threshold function	Collisions within the cluster are prevented	Energy efficiency per unit node has been improved	Better packet delivery rate	The initial state node is stationary
	EALC	[116]	Use a variant of the K-means density clustering algorithm for selection of cluster heads	Suitable for high-mobility networks	Network energy consumption has been reduced	Good cluster build time	Other performance metrics of the network were not considered
	CREEP	[117]	Modifying the CH selection thresholds in a two-level heterogeneous WSN	Heterogeneous network	Improved network lifetime	The scheme is extensible	Problems such as network latency were not considered
Data Link Layer	EL-MAC	[119]	Conserve energy through saving of communication cycles	Congestion-free, Collision-free	EE has been improved	Superior in terms of packet drop ratio, average throughput and end-to-end delay	Performance depends on location prediction accuracy
	EF-MAC	[120]	Clustering and reducing retransmission times	Suitable for time-critical sensing applications	Outperforms the conventional MAC protocols in terms of EE	Reduce network latency	Only considered one-dimensional data collection routing path
Cross Layer	EECRED	[121]	Transmission strategy	Cross-layer design	Reduced the energy consumption in two layers at the same time	The number of living nodes has increased	Other performance indicators such as delay were not taken into account

6.1. Replace the battery or UAV

In this subsection, the first two methods shown in the Fig. 5 are mainly introduced. In [123–126], the authors put forward the methods of replacing the depleted battery with a new battery or replacing the UAV with a fully charged battery, but those methods will incur many problems. First, these methods increase the cost of the whole UAV communication system because we need to have some spare batteries or UAVs on standby. Second, charging the battery or UAV requires a manual operation, which undoubtedly reduces the flexibility and intelligence of the UAV communication system. Although the authors in [127,128] designed the wireless contact charging method to charge the replaced UAV automatically, the coupling coil structure is optimized to improve the

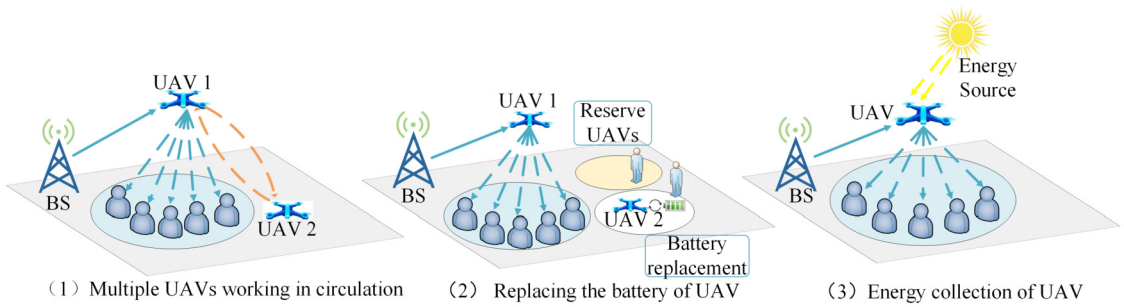

Fig. 5. Three methods for energy supplementation.

landing accuracy of UAV charging in [129], and communication is still interrupted when replacing the battery for the UAVs. Communication links need to be re-established with ground equipment and BSs for the UAVs newly added to the communication network. In summary, the above approach is not a sustainable and reliable solution, and EH has been recognized as a promising research approach in recent years. It refers to a permanent power supply technology that collect energy from the surroundings and convert it into electricity. This method not only charges working UAVs directly but also provides an effective endurance scheme for networks whose network life is affected by battery capacity limitations and that may be in dangerous environments where it is inconvenient to change batteries [130].

6.2. Energy harvesting technology

With the advocacy of environmental protection by society and policy, the utilization of clean and sustainable energy has been widely considered, and the UAV industry is no exception. Available EH sources of UAV communication mainly include solar energy, wind energy, heat energy, light energy and radiation energy [131]. It is worth mentioning that wind is a good alternative to solar radiation when the sun's intensity is too low to produce enough energy, and the two energy types tend to complement each other in time.

However, the application of this technology is not as simple as using an energy collector to collect energy. Unlike traditional power supply methods, UAV communication systems powered by EH must obey the constraint of the energy relationship, that is, the energy used before time T cannot exceed the energy harvested before time T [132]. On the other hand, due to the random fade of the channel, we cannot completely and accurately predict the energy required by UAVs for communication [133], which is undoubtedly a difficult problem that needs to be solved in the application of this technology.

Next, we will describe the methods by the different energy sources.

6.2.1. Environmental energy harvesting

Environmental energy mainly refers to the natural energy in nature, such as solar energy and wind energy. Solar energy collection is considered to be a very effective solution to overcome the energy limitation of UAVs. In [134], the feasibility of EH technology was verified, and two schemes were proposed. The first scheme aims to find the best balance between coverage and harvested energy, while the other seeks the balance between the flight time and the harvested energy of the UAV.

In [135] and [136], a solar UAV communication system was established. Authors in [135] considered the factors affected by clouds in the solar energy collection process of solar panels and analyzed the model of solar energy attenuation through clouds. As shown in Fig. 6, the optimal altitude range of a UAV flight was found, and the functional relationship of solar output power

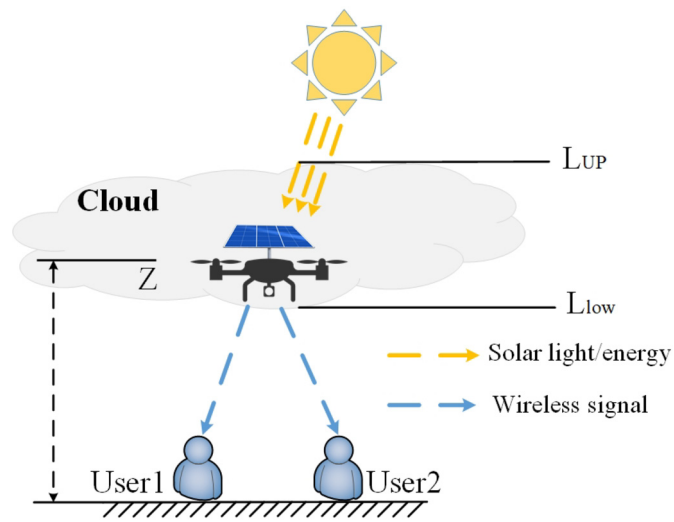

Fig. 6. Optimal altitude range of UAV flight.

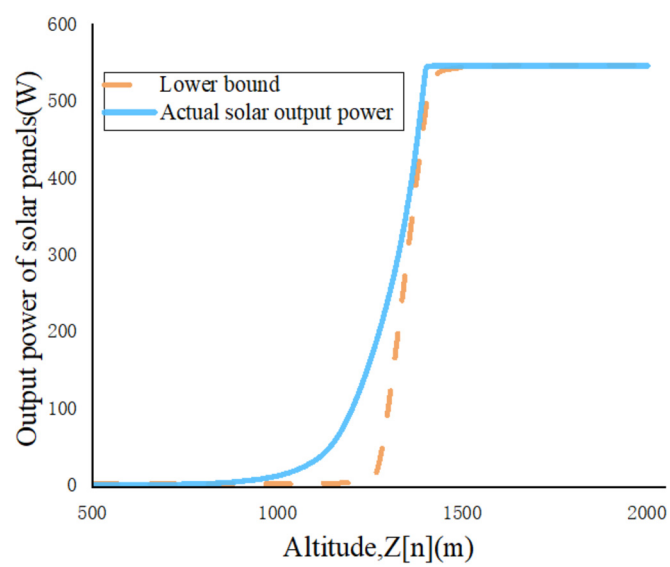

Fig. 7. Relationship between the solar output power and UAV height change.

changing with UAV flight altitude is established in Fig. 7. Therefore, the changing rule of the lower bound of solar output power is mastered, paving the way for us to carry out further trajectory planning and resource allocation. Authors in [133] realized the joint optimization of EH, trajectory design and resource allocation through power-cognitive technology.

In [136], the Poisson distribution is used to describe the whole process of a UAV collecting renewable energy. Under multiple com-

Table 7
Comparison of energy efficient methods in EH.

Methods	Article	Main idea	Features	Performance	Advantages	Limitations
Multiple UAVs working in circulation	[124–126]	Use fully charged UAV to replace the exhausted one	It can complete long endurance tasks	Automatic UAV replacement methods are provided	Provide the uninterrupted service to the users	Increase the cost of the whole UAV communication system
Replacing the battery of UAV	[123,127,128]	Change the batteries for the exhausted UAV	Low system flexibility and intelligence	Wireless charging of batteries is realized	Realize more rapid energy supplement	The battery replacement process may cause a system interruption
Energy harvesting of UAV	[133–140]	Environmental energy harvesting	The stability of the system is restricted by the speed of energy collection and consumption	Optimal altitude range of UAV flight	The energy is environmentally friendly and renewable	Limited by natural conditions
	[21,141–146]	RF energy harvesting	More suitable for the urban environment than others	Outage probability analysis	Less affected by the environment	Limited by the amount of RF energy in space
	[147–150]	Laser beam energy harvesting	Charging and communication occur simultaneously	Balances the tradeoff between maximizing the performance of laser energy harvesting versus wireless communication at the UAV	The beam can be used for both charging and data transmission	Only applicable to line-of-sight (LoS) communication

plex constraints, such as energy acquisition, multiple minimum rate outage probability (MMROP), average transmitting power (ATP) and average interference power (AIP), EE optimization targets are developed for UAV-based cognitive radio networks with energy harvesting (EH-UAV-CRN).

In the above articles, the natural environment is random. In the UAV group, because UAVs are in different environments, it can lead to an imbalance of energy harvested between different UAVs. Therefore, [137] is the first one to draw out the concept of energy cooperation. Through wireless transmission technology, the user can transmit part of the energy wirelessly to another energy acquisition user. This method can be applied to various wireless network applications. The studies of [138] and [139] are based on millimeter wave cellular networks. In [138], the UAV needs to consider whether the energy should be stored by itself or transferred to others in every time slot when it collects energy. In addition, the dynamic energy-aware power allocation (DEPA) algorithm is proposed to coordinate the transmitting power of UAVs, which maximizes the network utilization under the condition that the data and energy queue length are low. An optimized multiagent deep reinforcement learning algorithm is proposed in [139] to achieve an optimal allocation of the collected energy resources. Authors in [140] proposed two energy acquisition and information processing protocols, the base station-based time exchange protocol (TSR) and the base station-based right division protocol (PSR), and the outage probability and throughput of the two protocols were studied, proving that the PSR is superior to the TSR in both outage probability and throughput.

6.2.2. RF energy harvesting

In addition to collecting solar energy, electromagnetic (EM) radiation in the form of RF or microwave signals can also be utilized and then converted to power through rectifier circuits. Such RF sources could be the EM radiation from the surrounding environment or the beamforming signals emitted by the RF transmitters [131]. This resource is less affected than solar power by environmental factors such as weather. In this method, the UAV can directly receive energy without changing its position and flight trajectory, ensuring that the working of the UAV can operate as usual without interruption. Authors in [141] considered the efficiency of energy collection and the probability of interruption when RF transmitters are deployed on the ground in urban emergency scenarios. Authors in [142,143] minimized the energy transfer fading and communication interruption problems by optimizing the position of the RF transmitters. In [144], the wireless network is redesigned to make full use of the RF spectrum and radiation as well as the network infrastructure by integrating wireless power and wireless communication so that the dual purpose of communication and power supply had been achieved. In [21,145,146], the collected energy was not only used in UAVs but also extended to UAV-assisted IoT communication, solving the energy shortage problem of the IoT devices or the subordinate users by transmitting the collected energy to the IoT devices at the same time.

6.2.3. Laser beam energy harvesting

Laser power has also emerged as a viable solution to provide convenient and sustainable energy for UAVs. A laser-driven UAV wireless communication system is studied in [147], in which the laser beams are sent by the laser transmitter to charge the UAV in flight. In addition, the energy consumption of the UAV communication process is considered comprehensively, and the UAV trajectory and power distribution are jointly optimized to maximize the downlink communication throughput. Furthermore, authors in [148] studied the probability of interruption caused by laser charging with a UAV-based relaying system. Authors in [149] and [150] proposed a special method, combined with the long-distance transmission characteristics of free space optics (FSO), which equips UAVs with light collectors to implement the simultaneous transmission of energy and information between the backhaul communication from the base station to the UAV.

A summary of the existing methods for energy collection is presented in Table 7 and the main ideas, advantages and limitations of each method are briefly analyzed. At the beginning, we mentioned that the cost increase and communication interruption were the important problems faced by the UAV communication system. EH technology is the most effective way to solve the interruption problem. UAV communication technology is widely used, and different energy sources should be considered in different application

scenarios. But the trade-off between the speed of energy collection and the consumption is still a very critical problem to be solved, Contingency plans for emergencies caused by inadequate energy harvesting should also be considered.

In addition, the development of EH technology is also limited by the progress of related industries such as electromagnetic technology, the combination of EH technology with related optimization methods such as trajectory planning, resource allocation and protocol design will be of great benefit to solve the problem of cost increase and communication interruption.

7. Future research challenges

In the previous sections, the current energy saving technologies for UAVs were reviewed, with a preliminary understanding of the relevant technology. However, the application of these techniques has greatly improved the UAV energy consumption problem but did not go far enough to completely solve the problem. In-depth research is still needed in the future. Next, combined with the study and review of the former research, we put forward several potential problems that can be improved and further studied in the future to inspire the readers.

7.1. Accurate energy consumption model

The energy consumption model of the fixed-wing and rotary-wing UAVs has been introduced in the section of UAV trajectory planning and deployment simply. In addition to these two models, there should be many other energy consumption models [19]. As shown above, the energy consumption of UAVs is influenced by many factors, such as air resistance, gravity and so on, thus, the energy consumption model can not reflect the actual energy consumption. Moreover, intuitionistic energy consumption model is also needed to quantify the design of energy saving communication protocol, while joint feedback energy model is needed to energy harvesting and transfer to reflect the overall variation of energy consumption.

7.2. Holistic approaches and emerging techniques

As we showed above, most research has been directed towards a certain method for EE optimization. It is much difficult to achieve the desired thousand fold EE with a single approach. A holistic approach is thus necessary, i.e. energy efficient techniques are combined. Recently, UAV trajectory planning and resource allocation for communication resource are jointly optimized for EE [23,51]. Since different methods have different emphases and limitations, the key point of the combination is to find the junction or co-optimization variable, while the impact of limitations of different methods on other methods should be avoided. Furthermore, new emerging techniques can also be applied for energy efficient UAV communication. In particular, reconfigurable intelligent surface (RIS), caching and MEC. RIS can define new propagation and improve communication quality by adjusting the transmission environment of electromagnetic signals with low power consuming [151]. The deployment of caching can alleviate the backhaul transmissions, which is much important for UAV communication [152]. The application of MEC in UAV communication can alleviate the data-processing capabilities of UAVs. Whatever new technology, the basic problem is to establish the energy efficient problem model based on real application scenario.

7.3. The optimization of the UAV flying altitude

As shown in the existing research, many articles about the flight height optimization of UAVs optimize the height at a fixed value, but the height of UAVs changes dynamically during flight, where the optimization model can only for certain heights, and has no generality. However, higher altitude leads to larger path loss, which requires more energy to communicate with the ground users. Therefore, the variation of the energy consumption with flexible height is more realistic. Some scholars have recognized this, begun to combine flight height and horizontal position for energy consumption [48,49], but simple height transformation with two-dimensional coordinates attached are not able to reflect the variation of real time position of UAV, which are quite a lot of structures constraints and uncertainties. Thus, the flight height and the horizontal position could be combined to study the relationship between the UAV 3D dynamic position and EE. Unlike two dimensional (2D) path planning, the difficulties increase exponentially with dynamic constraints and kinematic constraints becoming much more complex of 3D. Furthermore, the optimization problem of flying altitude in 3D is usually a NP-hard problem, appropriate mathematical tools and optimization algorithms need to be carefully considered.

7.4. Reduce the propulsion energy consumption and communication energy consumption at the same time

UAV energy consumption includes communication energy consumption and propulsion energy consumption. Combined with the review of the relevant studies above, some researches only optimize the propulsion energy consumption or communication energy consumption while ignoring the impact on other parts of energy consumption of the overall system. Some authors have begun to optimize the total energy consumption of the two common parts in forms of a joint optimization in the literature of the past two years, which are simply solved by multi-objective optimization algorithm. Although communication energy consumption is much smaller than propulsion energy consumption, the optimization for them both have relationship with trajectory planning. However, the existing energy model can not reflect the proportionality of the two energy consumption parts. Therefore, the design of energy model and co-optimization problem are the first problem need to be solved for the co-optimization.

7.5. Trade off between energy and other performance

EE is only one of the key performance indicators for evaluating 6G wireless networks. Spectrum efficiency, peak data rate, latency, cyber security are also crucial for 6G networks. As that 6G will establish heterogeneous networks with integrating underwater (or sea) networks to form a large-dimensional space-air-ground-underwater network [153]. Thus multiple optimization of performance indicators is much important. Spectrum efficiency (SE) and EE are usually regarded as multiple optimization objectives, since they usually contradict with each other. The same as SE and EE, EE and cyber security are also two inherently contradictory factors. The more complicated the method is, the less chance the attacker can break down the network, while the more calculation resources are needed for sophisticated process, where more energy is consumed. Whatever performance indicators are optimized, the key issue is to establish the common optimization objectives and design appropriate optimization algorithm.

8. Conclusion

UAVs are expected to be an important component of 5G/6G that can potentially facilitate wireless broadcast. UAV communication is regarded as an important component to establish the space-air-ground-underwater networks, realize seamless integration of wireless information. Energy efficient communication is one of the

challenges for UAV communication due to the limitation of on-board energy, where a variety of energy efficient technologies have been proposed. In this article, existing energy efficient technologies were classified as four broad categories, at the same time, detailed analysis of main idea, outstanding features and limitations for each method were given. Finally, the open research issues and promising techniques are also given to inspire the readers. We hope that this paper and those in this issue will help to provide the summary and guiding significance for the follow-up work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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